Scoring and Aggregating Human Probability Forecasts on Two Public Corpora

An out-of-sample comparison of nine aggregation rules

Alejandro Guerrero Padrés

2026-08-16

Abstract

Given many people's probability forecasts on the same question, how should they be scored, and how should they be combined? This paper answers both questions against two public corpora of real human forecasts: the Good Judgment Project's ACE tournament surveys (2011–2015, 3.1 million individual-level rows) and the ForecastBench 2024 human round. Individual accuracy and calibration are measured under precisely stated protocols, and nine aggregation rules are then compared out of sample, with every free parameter tuned on the first two tournament years and frozen before evaluation on the last two and on the modern round. Aggregating is worth far more than choosing how to aggregate: on the held-out years the best rule scores 0.204 against 0.422 for the average member of the same crowd on the same days, while the spread across rules is 0.086. Log-odds extremizing wins on the tournament data and keeps improving as the crowd grows, where the simple mean saturates near twenty-five forecasters. It does not travel: the same constant ranks seventh of eight on ForecastBench's public crowd, which is overconfident where the tournament's independents are underconfident. Extremizing constants are properties of a forecaster pool rather than of aggregation in general. The scored ForecastBench set reproduces the benchmark leaderboard's 521 question-horizons exactly, and the residual gap against its published scores is traced to the leaderboard's difficulty adjustment rather than left open.

1 Introduction

A group of people gives probabilities for the same event. Some are careful, some are not; some are confident, some hedge. Two questions follow, and they are usually run together when they should be kept apart. The first is how to score what each person said, which is a question about proper scoring rules and about the accounting conventions that turn a stream of revised forecasts into a number. The second is how to pool them, which is a question about aggregation rules and about whether a rule that looks good on one pool of forecasters keeps working on another.

The wisdom-of-crowds literature has good answers to the second question and they are not all compatible. Averaging probabilities is the obvious rule and a poor one: it pulls the aggregate toward the middle regardless of how many people agree. Pooling in log odds does better. Pushing the pooled log odds further from 50% — extremizing — does better still on tournament data, with two standard justifications: the crowd's information is partly independent, so the pooled forecast should be more confident than any member (Baron et al. 2014), and the standard aggregate is systematically underconfident under a shared-signal model (Satopää et al. 2014). Trimming the tails is a competing family (Powell et al. 2024), and extremizing by a factor derived from the crowd size rather than a fitted constant is a third (Neyman and Roughgarden 2022). Selecting a subcrowd on past accuracy is a fourth (Mellers et al. 2014).

What is harder to find is a clean statement of how much any of this is worth relative to the decision to aggregate at all, measured against a baseline drawn from exactly the same population on exactly the same days, with every free parameter frozen before the evaluation data is touched. That is what this paper reports. Two corpora are used, thirteen years apart, so that a rule tuned on the first can be checked against the second without re-tuning.

Three findings. Aggregating is the larger decision by a factor of about two on both non-expert crowds, but not by an order of magnitude — the choice of rule earns its keep, it just earns less than the choice to pool. Extremizing wins on the tournament data and keeps gaining as the crowd grows, exactly where the plain mean stops. And it does not transfer: the constant tuned on the tournament's independent forecasters is near the back of the field on ForecastBench's public crowd, for a reason visible in the calibration diagrams of both.

Everything here is reproducible from a public repository: raw files verified by checksum, a build that fails on any of 16 hard data invariants, and 352 unit-test assertions. Section 6 gives the details.

2 Data

2.1 Good Judgment Project ACE surveys

The primary corpus is the Good Judgment Project's forecast archive from IARPA's ACE tournament, four seasons between 2011 and 2015, published on Harvard Dataverse under CC0 (Good Judgment Project 2016). This analysis uses the question file and the four survey-forecast files: 3,143,460 rows, one per answer option per forecast event, with no exact duplicates.

The tournament assigned participants to elicitation conditions, and this matters more than any other structural fact about the corpus. Condition group 1 forecast independently on a survey platform. Group 2 could see crowd information. Group 4 worked in interacting teams. Group 5 were superforecaster teams, formed after year 1 by promoting top performers. Group 3, the prediction markets, appears only in separate market files and not in the survey data at all. Team and superteam members are not independent crowd members — they talk to each other — so the **primary population throughout is condition group 1**, with the other groups reported only as comparison cohorts. Training subconditions within group 1 are kept as covariates rather than filtered out.

The analysis question set is the closed, non-voided, regular binary questions: 303 of the 617 in the file. Binary questions make proper-score comparisons across aggregation rules clean, since each forecast is a single probability. The exclusions are counted, not silent: 119 voided, 84 ordered multinomial, 79 conditional branches, and 32 unordered multinomial questions, one stated reason each. Ordered questions would need a distance-aware score; conditional branches score only under the realized condition.

Two format quirks are handled at ingest and recorded because they would otherwise corrupt the question table silently: the question file uses classic-Mac carriage-return line endings with Latin-1 text, converted at the byte level before any string handling, and `m/d/y` dates. One structural fact is worth stating because it is what makes trailing-accuracy methods possible at all: returning participants keep their identifiers across seasons, so a forecaster's history spans years even though the codebook's identifier ranges are banded by recruitment year.

cohort	scored events	forecasters	questions
independent individuals	314,372	6,473	303
teams	198,233	2,592	303
superteams	99,976	181	233
saw crowd information (yr 1)	45,816	657	94
top decile (yrs 2-4)	2,532	27	181

Table 1: GJP cohorts on the analysis question set. Rows are last-per-day forecast events. The top decile is a subset of the independents, selected on year-1 accuracy alone and labelled only from year 2 onward.

2.2 ForecastBench 2024 human round

The validation corpus is the human round of ForecastBench, a continuously updated forecasting benchmark (Karger et al. 2025), taken from its public dataset repository at a pinned commit under CC BY-SA 4.0 (Forecasting Research Institute 2024). The round posed 200 questions on 2024-07-21 to two comparison groups: a public crowd (500 forecasters) and a superforecaster group (39).

The structural fact that governs the whole layer is that **a forecaster answers each question once per resolution horizon**. The round sets eight horizon dates running from one week to ten years out, and each forecast entry carries its own. Horizon is therefore part of the forecast key, not a property to be averaged over.

Only dataset-sourced questions are scored. ForecastBench resolves market-sourced questions that are still open at a horizon against the market price on that date, which is a continuous proxy rather than an outcome, so including them would mix two different quantities under one score. Of the 200 questions, 110 are dataset-sourced (ACLED, DBnomics, FRED, Wikipedia, Yahoo Finance) and 105 of those carry a resolution; 5 carry none at all.

2.3 Exclusions

Both corpora are reduced by exclusions, and in a study whose conclusions are differences of a few hundredths, an uncounted exclusion is a real hazard: it is exactly how a population quietly stops matching its baseline. Every drop is therefore a counted row, and two hard build checks require the counts to close against the tables they describe. A filter added without a reason breaks the identity and fails the build.

The scored ForecastBench set is 29,599 rows over 521 question-horizons. On the GJP side, 1,739,566 of the 3,143,460 source rows sit on questions outside the analysis set, leaving 1,403,894 accepted rows forming 701,947 paired forecast events.

3 Methods

3.1 Scores

Scores are the two-option Brier score (Brier 1950): for a binary question with forecast p and outcome $y \in \{0, 1\}$, $\mathrm{BS} = (p - y)^2 + ((1 - p) - (1 - y))^2 = 2(p - y)^2$, ranging from 0 for a perfect forecast to 2 for a confident wrong one. This is the convention of the tournament literature the results are reconciled against; where an external anchor uses the single-probability convention $(p - y)^2$

stage		entries
source	all human-round entries	55,372
	no resolution date (market-sourced questions)	5,096
	probability outside [0, 1]	281
	duplicate (forecaster, question, horizon) keys collapsed	0
kept	horizon entries carried forward	49,995
	question never resolved upstream	2,368
	horizon not covered by the resolution set	18,028
scored	scored rows	29,599

Table 2: ForecastBench exclusion ledger. Each block sums to the line above it. The second block is what makes this corpus a snapshot rather than a history: the round's resolution set reaches five of the eight horizons, so forecasts aimed at 2027, 2029 and 2034 have nothing yet to score against.

the table says so. The Brier score is strictly proper (Gneiting and Raftery 2007), so a forecaster minimises it by reporting their true belief. Log scores clamp probabilities to $[10^{-3}, 1-10^{-3}]$.

Calibration is reported as reliability diagrams over ten equal-width bins with Wilson intervals on the observed frequency in each bin (Wilson 1927), and as the Murphy decomposition of the mean score into reliability, resolution and uncertainty (Murphy 1973), computed on the same binning so that the identity holds exactly against the binned mean.

3.2 The two scoring protocols

The corpora are shaped differently and are given different protocols; the difference is stated rather than smoothed over.

GJP: daily carry-forward. A forecast stands from the day it is submitted until the day before that forecaster's next forecast on the question, or the end of the scoring window, whichever comes first. The scoring window runs from the question's start date through its suspension date, falling back to its close date. A forecaster's score on a question is the day-weighted mean Brier of their standing forecasts; scores are then averaged across questions without weighting, so that a question open for 550 days does not count for more than the median question's 78. Edge rules: multiple forecasts on one day resolve to the last by timestamp; forecasts entered before the window carry from its start; forecasts after it contribute zero days; a withdrawal is scored on its own day using the forecaster's last standing values, after which they leave the question until any later forecast re-enters.

The scoring cell is (question, forecaster, condition code, tournament year) rather than (question, forecaster). Participants were re-randomised between seasons, so on the 31 questions that straddle a year boundary a forecaster can appear under two experimental conditions on the same question — 3,730 such cases. Keying on the condition code keeps each cohort's days its own, which is what allows a crowd to be compared against exactly the forecasters in it.

ForecastBench: single shot per horizon. Each forecast is keyed by forecaster, question and horizon date, and no forecast is revised. A crowd is pooled per question-horizon, and question-horizons average unweighted within a cohort.

rule	definition	source
mean	arithmetic mean of the probabilities	–
median	median of the probabilities	–
trimmed mean (40%)	mean after dropping equal counts from each tail	–
HD-trim (40%)	mean inside the shortest interval holding 1 - p	Powell 2024
softened mean (30%)	trims only the tail the mean leans toward	aggutils
geometric mean of odds	mean taken in log odds, mapped back	–
extremized (a = 2)	pooled log odds multiplied by a	Satopaa 2014
Neyman extremized	pooled log odds times a crowd-size factor	Neyman 2022
select crowd (top 10)	median of the top k by trailing accuracy	Mellers 2014

Table 3: The nine aggregation rules. Parameterised rules show the value frozen after tuning. Rules that work in log odds clamp probabilities to the same bounds as the log score.

3.3 The aggregation rules

Nine rules, listed in Table 3. Four delegate to `aggutils` (Hickman and Jacobs 2023), the reference implementation of the trimming and extremizing methods; that package works on the 0–100 percentage scale, so the wrappers convert on the way in and out. The other five are implemented directly. Unit tests cross-check every wrapper against an independent reimplementation — the symmetric trimmed mean at every crowd size from 1 to 12, the Neyman crowd-size factor against its published formula, and the highest-density trimmed mean against a shortest-interval implementation — so the package is used as a reference rather than trusted blindly.

Two of these need a word. The extremized rule with $a = 1$ is exactly the geometric mean of odds, so the family contains its own null hypothesis and the tuned value can be read directly as how much extremizing helps. The Neyman rule takes its factor from the crowd size rather than from a fitted constant, which means it has nothing to tune and nothing to transfer — a property worth testing separately from whether extremizing itself transfers.

3.4 The day panel and the average-member baseline

The headline comparison is a crowd against its own members, so both have to be scored the same way. For GJP the panel makes that exact: the scoring segments of the primary population are expanded into one row per (question, forecaster, day) on which that forecaster had a standing forecast — 12,933,059 rows over 33,204 question-days. On each day the rule turns the standing forecasts into one probability and that probability is scored; the question's score is the mean over its days.

The average-member baseline runs on the same panel: on each day the standing forecasters are scored individually and averaged, then days average as they do for every rule. Same population, same days, same weighting. A hard build check requires the panel to hold exactly the days the protocol scores, forecaster by forecaster, which is what keeps the comparison honest.

Crowds are large. On the evaluation years a question-day carries a median of 349 standing forecasters, ranging from 8 to 3,468.

3.5 Tuning, freezing, and the leakage rule

Every free parameter — the two trim fractions, the softening fraction, the extremizing constant, and the select-crowd size — is chosen on **GJP years 1–2** and then frozen. Evaluation is **GJP years 3–4**, never seen during tuning, plus the ForecastBench round as external validation. A question belongs to the year it was first forecast in, so a question straddling a boundary lands on one side only. The split holds 148 tuning questions against 155 evaluation questions.

Two structural guards, rather than care, keep the split intact. Every tunable parameter is declared in one grid function and nowhere else, so the tuning stage cannot silently acquire a knob. And the tuner raises an error if it is handed scores carrying an evaluation year, which a unit test exercises.

Select-crowd needs a third guard, because it is the one rule that looks at the past. A forecaster's trailing accuracy on a given day is a running mean over the questions that finished scoring **strictly before** that day, joined as of the forecast date, and a forecaster needs 5 finished questions before becoming eligible. Where fewer than two forecasters qualify the rule falls back to the whole-crowd median and the fallback days are counted. On the evaluation years the fallback rate was 0.0%. Select-crowd is GJP-only: a single ForecastBench round carries no history to select on, and the results table says so rather than leaving an unexplained gap.

3.6 Uncertainty

All intervals are question-cluster bootstraps: question identifiers are resampled with replacement from a table of precomputed per-question summaries, 2,000 replicates, and the statistic is recomputed each time (Efron and Tibshirani 1993). Clustering on the question is the right unit because forecasts within a question share an outcome. For ForecastBench the cluster is the question, so the several horizons of one question resample together. Nothing upstream of the summaries is re-run per replicate, which is what keeps 2,000 replicates affordable over a twelve-million-row panel.

4 Results

4.1 Individual accuracy and calibration

Table 4 gives the descriptive picture the aggregation results build on. The cohort ordering matches the tournament literature — superteams ahead of teams ahead of independent individuals — and the year-1 top decile stays well ahead of the pool it was drawn from in later years, which is the basic evidence that forecasting skill is a persistent trait rather than a run of luck (Mellers et al. 2014; Tetlock and Gardner 2015).

Figure 1 shows two distinct failure modes, and the distinction is what the transfer result in Section 4.5 turns on. The ForecastBench superforecasters track the diagonal closely. The public crowd flattens above 50%: high probabilities are not matched by outcomes, which is overconfidence. The GJP cohorts sit below the diagonal across the upper half in the opposite direction — they overestimate the probability of the non-status-quo option, which is the familiar overprediction of change, and in this question set the status quo won 224 of 303 times.

The Murphy decomposition in Table 5 makes the same point numerically and adds one. Miscalibration is a modest share of every cohort's score; what separates the strong cohorts is resolution — they place probabilities nearer the extremes and are right about it. The ForecastBench public crowd is the exception in both terms at once: the largest reliability penalty in the table and the smallest resolution.

cohort	n	averaged over	mean Brier [0-2]
ForecastBench public	25,716	scored rows	0.468
GJP independent individuals	303	questions	0.384
GJP teams	303	questions	0.293
GJP top decile (yrs 2-4)	181	questions	0.247
ForecastBench superforecasters	3,883	scored rows	0.229
GJP superteams	233	questions	0.168

Table 4: Individual accuracy. GJP scores are day-weighted per forecaster and question, averaged unweighted across questions. ForecastBench pools scored rows within a cohort.

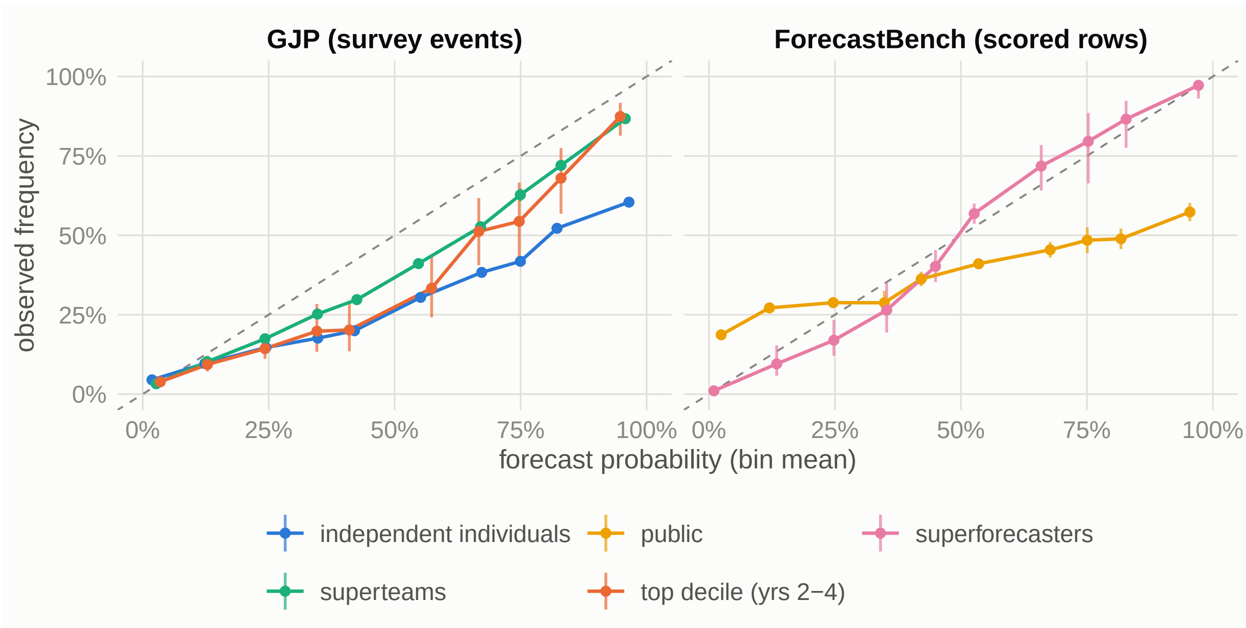

Figure 1: Reliability diagrams, ten equal-width bins, Wilson 95% intervals on the observed frequency. The dashed line is perfect calibration. Points below the line are forecasts that were too high.

corpus	cohort	reliability	resolution	uncertainty	Brier (binned)
ForecastBench	public	0.058	0.025	0.436	0.468
ForecastBench	superforecasters	0.003	0.182	0.411	0.231
GJP	independent individuals	0.074	0.067	0.338	0.345
GJP	superteams	0.010	0.149	0.358	0.219
GJP	top decile (yrs 2-4)	0.018	0.109	0.303	0.212

Table 5: Murphy decomposition, event-pooled, two-option scale: Brier = reliability - resolution + uncertainty. Lower reliability is better (it is the miscalibration penalty); higher resolution is better.

family	grid searched	frozen	tuning-year Brier
extremized	1, 1.25, 1.5, 2, 2.5, 3	2.0	0.235
hd_trim	0.05, 0.1, 0.2, 0.3, 0.4, 0.5	0.4	0.271
select_crowd_k	5, 10, 20, 40, 80	10.0	0.233
soften_mean	0.05, 0.1, 0.2, 0.3, 0.4	0.3	0.274
trimmed_mean	0.05, 0.1, 0.2, 0.3, 0.4, 0.45	0.4	0.297

Table 6: Parameters chosen on GJP years 1-2 and frozen before any evaluation data was scored.

4.2 Tuning

Table 6 gives the frozen parameters. Every one is an interior minimum of its grid; the first grid put three families on a boundary and was widened once, before any evaluation-year data was read. Extremizing has a clear optimum at a = 2: pushing the pooled log odds outward helps, and past that point the crowd becomes overconfident. The trimming families keep improving until they discard roughly 40% from each side, which is close to behaving like a median — consistent with trimming's usual role of removing confident tails rather than reshaping the middle.

4.3 What the crowd is worth

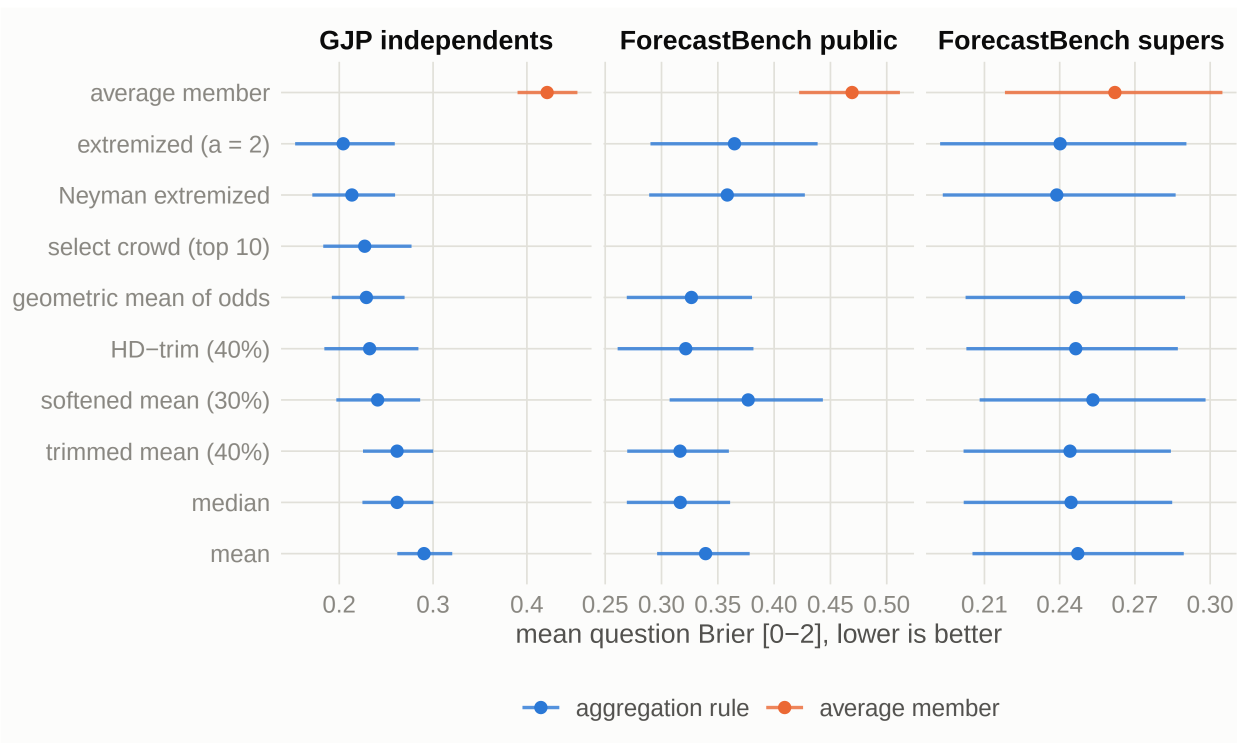

Figure 2: Aggregation rules out of sample, with 95% question-cluster bootstrap intervals. The GJP panel is the held-out years 3-4. Parameters were frozen on GJP years 1-2. The three panels are different question sets and are not comparable across panels; what is comparable within a panel is each rule against the average member of the same crowd, in orange. Select crowd runs on GJP only.

On the held-out GJP years the best rule scores 0.204 against 0.422 for the average member of the

paired difference in mean Brier	estimate	2.5%	97.5%
GJP years 3-4: average member - extremized (a = 2)	0.217	0.186	0.246
ForecastBench public: average member - trimmed mean (40%)	0.153	0.142	0.164
ForecastBench supers: average member - Neyman extremized	0.023	0.013	0.032

Table 7: Paired per-question differences between the average member and the best rule for that cohort, with question-cluster bootstrap intervals. Positive means the crowd is better.

same crowd on the same days — a reduction of 52%. Most of it needs no cleverness. The weakest rule in the field, the plain mean, already reaches 0.290; going from there to the best rule is worth a further 0.086. Measured as the step from the average member to the weakest rule against the spread among the rules, aggregating is the larger decision by a factor of 1.5 — not by an order of magnitude.

The same ratio is 1.5 on the ForecastBench public crowd and 0.6 on the superforecasters. The supers are the interesting case: the cohort is close enough to its own aggregate that the rules are spread further apart than the aggregate is from a typical member, and every one of those gaps is smaller than its own interval. Aggregation buys the most where the individuals disagree the most. That is the mechanism, not an incidental finding, and it is the reason a rule comparison run only on expert crowds would understate what aggregation does for everyone else.

4.4 Crowd size

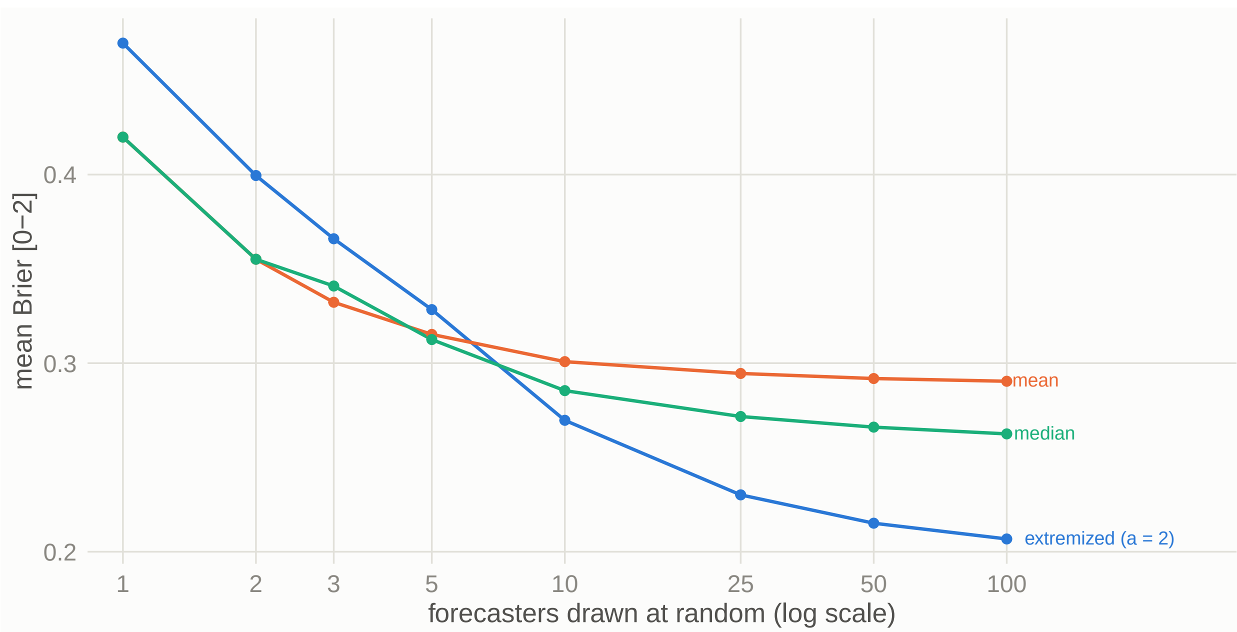

Figure 3: Accuracy against crowd size on the GJP evaluation years. On each scoring day, up to n of that day's standing forecasters are drawn, so every size is scored on the same days; 5 draws per question and size, up to 30 days per question. At n = 1 the mean and median reduce to the drawn forecast and land on the average-member baseline.

Two things separate the rules in Figure 3. The simple mean flattens early: past roughly 25 forecasters it stops improving, because averaging probabilities pulls the aggregate toward the middle no matter how many people agree. Extremizing keeps gaining, because a larger crowd both cancels more noise and justifies more confidence in the pooled signal. At one forecaster the extremized rule is the worst of the three, which is the same fact seen from the other side: there is nothing to be confident about in a single opinion.

The $n = 1$ point is also the design check. Drawing the subcrowd from each day's standing forecasters, rather than once per question, is what keeps the curve about size: a crowd sampled once per question covers more of a question's hard early stretch as it grows, and that extra coverage would show up as a size effect that is really a coverage effect. Because the draw is per day, every size is scored on the same days, and at $n = 1$ the mean and median land on 0.420 against 0.422 for the average member computed independently in Table 7 — a residue of 0.002 left over from thinning the days.

4.5 The extremizing constant does not transfer

The constant tuned on GJP is worth 0.086 against the plain mean there, and costs accuracy on the modern round. On the ForecastBench superforecasters the extremized rule is near the front and near-identical to untuned pooling in log odds. On the public cohort it finishes 7th of 8, behind even the plain mean, while the trimmed mean (40%) wins.

This is Figure 1 arriving from the other direction. The GJP independents are collectively underconfident, so pushing their pooled forecast outward helps. The ForecastBench public crowd is already overconfident above 50%, and extremizing an overconfident crowd makes it worse. An extremizing constant is a property of a forecaster pool, not of aggregation in general, and one carried across pools without first checking the pool's calibration is a liability rather than a free improvement.

The Neyman rule is the natural control, since it derives its factor from crowd size and has no constant to carry. It is spared the transfer question but not the underlying one: it still extremizes, so while it is the best rule on the supers it finishes 6th of 8 on the public — one place ahead of the tuned constant, not in a different class. What fails to transfer is extremizing itself, on a pool that does not need it.

4.6 Reconciliation against published values

A reproduction that agrees with nothing published is a bug until proven otherwise, so the results are anchored in three places.

The scored set contains exactly the leaderboard's 521 dataset question-horizons. The reproduced medians sit about 0.008 below the published figures, and the reason is that the two are not the same statistic. The leaderboard's Brier column is difficulty-adjusted: ForecastBench fits a two-way fixed-effects model of the score on question and model, subtracts each row's estimated question effect, and rescales so that an always-0.5 forecaster averages 0.25. Because the human entries cover only this one round while the field of entrants spans many, the question effects do not average out for them and the adjustment lands as a roughly constant offset. The quantity that survives the adjustment is the gap between the two human cohorts, and it does: 0.036 here against 0.034 published. Nothing was tuned to close the difference, and the raw figures are what this paper reports throughout.

Absolute GJP values are not chased precisely, because published figures for that tournament vary with standardisation, year window and question set. The anchors used instead are the cohort

cohort	rule	mean Brier [0-2]	95% interval
ForecastBench public	trimmed mean (40%)	0.316	[0.270, 0.360]
ForecastBench public	median	0.317	[0.269, 0.361]
ForecastBench public	HD-trim (40%)	0.321	[0.261, 0.382]
ForecastBench public	geometric mean of odds	0.327	[0.269, 0.380]
ForecastBench public	mean	0.339	[0.296, 0.378]
ForecastBench public	Neyman extremized	0.358	[0.289, 0.427]
ForecastBench public	extremized (a = 2)	0.365	[0.290, 0.439]
ForecastBench public	softened mean (30%)	0.377	[0.307, 0.443]
ForecastBench public	average member	0.469	[0.422, 0.512]
ForecastBench supers	Neyman extremized	0.239	[0.193, 0.286]
ForecastBench supers	extremized (a = 2)	0.240	[0.192, 0.291]
ForecastBench supers	trimmed mean (40%)	0.244	[0.202, 0.284]
ForecastBench supers	median	0.245	[0.202, 0.285]
ForecastBench supers	HD-trim (40%)	0.246	[0.203, 0.287]
ForecastBench supers	geometric mean of odds	0.246	[0.202, 0.290]
ForecastBench supers	mean	0.247	[0.205, 0.289]
ForecastBench supers	softened mean (30%)	0.253	[0.208, 0.298]
ForecastBench supers	average member	0.262	[0.218, 0.305]
GJP independents	extremized (a = 2)	0.204	[0.153, 0.259]
GJP independents	Neyman extremized	0.213	[0.171, 0.259]
GJP independents	select crowd (top 10)	0.227	[0.183, 0.277]
GJP independents	geometric mean of odds	0.229	[0.192, 0.270]
GJP independents	HD-trim (40%)	0.232	[0.184, 0.284]
GJP independents	softened mean (30%)	0.241	[0.197, 0.286]
GJP independents	trimmed mean (40%)	0.262	[0.225, 0.300]
GJP independents	median	0.262	[0.225, 0.300]
GJP independents	mean	0.290	[0.262, 0.320]
GJP independents	average member	0.422	[0.390, 0.454]

Table 8: Full results. Rows within a cohort are comparable; rows across cohorts are different question sets. The GJP rows cover 155 held-out questions; the ForecastBench rows cover 105 questions over 521 question-horizons. Intervals are 95% question-cluster bootstraps over 2,000 replicates.

anchor	published	this paper
FB leaderboard: supers, median forecast	0.131 (adjusted)	0.122
FB leaderboard: public, median forecast	0.165 (adjusted)	0.158
FB leaderboard: gap between the cohorts	0.034	0.036
Probability-training gain (Mellers et al. 2014)	about 10%	7.6% (1a 0.398, 1b 0.368)
Tournament cohort ordering	supers < teams < independents	0.168 < 0.293 < 0.382

Table 9: Reconciliation. The ForecastBench rows use the single-probability convention to match the published leaderboard; everything else in this paper is on the two-option scale.

ordering, the probability-training effect, and the exact ForecastBench question-horizon match, all of which reproduce.

5 Limitations

Selection into the corpora. Both populations volunteered for forecasting tournaments. Nothing here describes how an arbitrary group of people would perform, and the crowd-size curve in particular is a curve over this pool.

The population shifts across the split. The evaluation years are not the tuning years with more data. The tuning years hold 1,212 forecasters against 5,698 in the evaluation years, superforecaster teams were extracted after year 1, and conditions were added and dropped between seasons. This is the honest version of an out-of-sample test on tournament data — a temporal split carries a population change with it — and it means the transfer result inside GJP is a weaker claim than the transfer result across corpora.

Binary questions only. Ordered and conditional questions are excluded with counts. A distance-aware score would be needed for the former and a realized-condition protocol for the latter, and neither is in scope here.

One modern round. ForecastBench contributes a single round, at five scored horizons out of eight. It cannot support forecaster histories, so select-crowd does not run there, and the transfer conclusion rests on one snapshot of two cohorts rather than on a series.

Aggregate versus individual comparison. The average-member baseline is what a randomly chosen member of the standing crowd scores, which is the right comparison for "is the crowd worth more than one of its members". It is not the same as the best member, who cannot be identified in advance; the select-crowd rule is the closest this paper comes to that question and it finishes mid-field.

The crowd-size curve is thinned. Days are subsampled to at most 30 per question to keep the sweep affordable. The residue this leaves is visible and quantified in Section 4.3: about 0.002 at $n = 1$.

6 Reproducibility

The analysis is R end to end (R Core Team 2026), over a local DuckDB database (Raasveldt and Mühleisen 2019), with dependencies locked by `renv` and continuous integration on every push. `make all` runs the chain from verified raw files to results: checksum verification, ingest, the two scoring layers, the aggregation experiments, the QA gate, and the tests. Raw data is never committed; the manifest of file sizes and MD5 sums is, and no build starts until every file matches it.

Three properties are worth stating because they were not free. The build fails on any of 16 hard invariants, including the two accounting identities of Section 2 and the requirement that the day panel hold exactly the days the protocol scores. The suite is 352 assertions over hand-computed fixtures, including the segment arithmetic, the strict leakage boundary, and a cross-check of every aggregation rule against an independent implementation. And the results are reproducible run to run: SQL grouping promises no row order, and both the bootstrap and the crowd-size draw read row order, so every read that feeds them is explicitly ordered — without which identical data produced different intervals on repeated runs.

Repository: https://github.com/nacapule/crowd-forecast-evaluation.

References

Baron, Jonathan, Barbara A. Mellers, Philip E. Tetlock, Eric Stone, and Lyle H. Ungar. 2014. "Two Reasons to Make Aggregated Probability Forecasts More Extreme." *Decision Analysis* 11 (2): 133–45. https://doi.org/10.1287/deca.2014.0293.

Brier, Glenn W. 1950. "Verification of Forecasts Expressed in Terms of Probability." *Monthly Weather Review* 78 (1): 1–3. https://doi.org/10.1175/1520-0493(1950)078%3C0001:VOFEIT%3E2.0.CO;2.

Efron, Bradley, and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman; Hall.

Forecasting Research Institute. 2024. *forecastbench-datasets*. Https://github.com/forecastingresearch/forecastbench-datasets.

Gneiting, Tilmann, and Adrian E. Raftery. 2007. "Strictly Proper Scoring Rules, Prediction, and Estimation." *Journal of the American Statistical Association* 102 (477): 359–78. https://doi.org/10.1198/016214506000001437.

Good Judgment Project. 2016. *GJP Data*. Version V1. Harvard Dataverse. https://doi.org/10.7910/DVN/BPCDH5.

Hickman, Molly, and Zachary Jacobs. 2023. *aggutils: Utilities for Aggregating Probabilistic Forecasts*. https://CRAN.R-project.org/package=aggutils.

Karger, Ezra, Houtan Bastani, Chen Yueh-Han, et al. 2025. "ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities." *Proceedings of the 13th International Conference on Learning Representations (ICLR 2025)*. https://iclr.cc/virtual/2025/poster/28507.

Mellers, Barbara, Lyle Ungar, Jonathan Baron, et al. 2014. "Psychological Strategies for Winning a Geopolitical Forecasting Tournament." *Psychological Science* 25 (5): 1106–15. https://doi.org/10.1177/0956797614524255.

Murphy, Allan H. 1973. "A New Vector Partition of the Probability Score." *Journal of Applied Meteorology* 12 (4): 595–600. https://doi.org/10.1175/1520-0450(1973)012%3C0595:ANVPOT%3E2.0.CO;2.

Neyman, Eric, and Tim Roughgarden. 2022. "Are You Smarter Than a Random Expert? The Robust Aggregation of Substitutable Signals." *Proceedings of the 23rd ACM Conference on Economics and Computation (EC '22)*, 990–1012. https://doi.org/10.1145/3490486.3538243.

Powell, Ben, Ville A. Satopää, Niall MacKay, and Philip E. Tetlock. 2024. "Skew-Adjusted Extremized-Mean: A Simple Method for Identifying and Learning from Contrarian Minorities in Groups of Forecasters." *Decision* 11 (1): 173–93. https://doi.org/10.1037/dec0000191.

R Core Team. 2026. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.

Raasveldt, Mark, and Hannes Mühleisen. 2019. "DuckDB: An Embeddable Analytical Database." *Proceedings of the 2019 International Conference on Management of Data (SIGMOD '19)*. https://doi.org/10.1145/3299869.3320212.

Satopää, Ville A., Jonathan Baron, Dean P. Foster, Barbara A. Mellers, Philip E. Tetlock, and Lyle H. Ungar. 2014. "Combining Multiple Probability Predictions Using a Simple Logit Model." *International Journal of Forecasting* 30 (2): 344–56. https://doi.org/10.1016/j.ijforecast.2013.09.009.

Tetlock, Philip E., and Dan Gardner. 2015. *Superforecasting: The Art and Science of Prediction*. Crown.

Wilson, Edwin B. 1927. "Probable Inference, the Law of Succession, and Statistical Inference." *Journal of the American Statistical Association* 22 (158): 209–12. https://doi.org/10.1080/01621459.1927.10502953.